

Validation of the Copenhagen Hip and Groin Outcome Score (HAGOS) using modern test theory across different cultures and languages: a cross-sectional study of 452 male athletes with groin pain

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ABSTRACT

Background No studies have tested the validity of the Copenhagen Hip and Groin Outcome Score (HAGOS) using modern test theory, across different cultures and languages.

Objective To validate the Danish, English and Norwegian versions of HAGOS and its six subscales (Symptoms (S, Item 1–7), Pain (P, Item 1–10), activities of daily living (Item 1–5), Sport and recreation (Sport/rec, Item 1–8), Participation in physical activity (item 1–2) and quality of life (item 1–5)) by evaluating differential item functioning (DIF) and measurement invariance across the three language versions in male multidirectional team athletes with groin pain. Second, to modify subscales depending on goodness-of-fit to the item response theory models and calculate conversion tables if language DIF was observed.

Methods We included individual responses to the Danish (n=157), English (n=146) and Norwegian (n=149) language versions of HAGOS from 452 athletes (median age 24 years old, range 20–28) with groin pain. Overall fit, model fit, individual item fit, local response dependence and measurement invariance was examined using confirmatory factor analysis and graphical Rasch models.

Results The removal of seven misfitting items (S2, P1, P2, A4, SP1, SP5, Q3) resulted in 6 HAGOS subscales with acceptable psychometric properties. For the Symptoms, Pain and Sports subscales evidence of DIF was disclosed between the three different language-versions of HAGOS and conversion tables were created.

Conclusions A revised HAGOS derived using modern test theory provides valid measurements for male multidirectional athletes with groin pain across different cultures and languages. Conversion tables must be applied to compare HAGOS scores from Danish, Norwegian and English language versions.

and recreation; participation in physical activity (PA) and quality of life (QOL).

The HAGOS was validated following the CONsensus-based Standards for the selection of health Measurement Instruments checklist using Classical Test Theory (CTT) and the original paper was published in 2011,¹ but no subsequent validation using modern psychometric methods has been published.² Furthermore, since 2011, HAGOS has been translated in more than 15 different languages (www.koos.nu), however, formal evaluation of the cross-cultural validity has not been studied. Data obtained using HAGOS in different countries are increasingly compared although such discussions are based on the assumption that the same construct is being measured across different language versions of HAGOS. This suggests that a subscale score of, say, 50 points in one HAGOS language version always translates to a HAGOS score of 50 points across other language versions. This is the assumption of measurement invariance, which is something that should be tested for gender groups, age groups and language groups. Here, we focus on measurement invariance across language groups.

The gold standard for validation of patient reported outcomes and for the study of measurement invariance is modern test theory (MTT), a collection of statistical models including confirmatory factor analysis (CFA),³ item response theory (IRT)⁴ and Rasch analysis.^{5,6}

The aim of this study was to test the construct validity of the HAGOS using MTT. Furthermore, measurement invariance and differential item functioning (DIF) across language versions were evaluated, and conversion tables between the three versions of HAGOS (Danish, Norwegian and English language) were presented where relevant and possible.

INTRODUCTION

The Copenhagen Hip and Groin Outcome Score (HAGOS) questionnaire was developed to obtain a quantitative measure of hip and groin disability in young to middle-aged physically active individuals.¹ HAGOS uses the International Classification of Functioning, Disability and Health framework and consists of six separately scored subscales: Symptoms; Pain; activities of daily living (ADL); Sport

METHODS

We sampled a total of 452 multidirectional male team-sport (football) athletes with current groin pain, confirmed with positive hip adduction squeeze test on examination,^{7–10} 18–35 years of age, who had filled out the Danish (N=157, 34.7%), English (N=146, 32.2%) or Norwegian (N=149, 33.0%) version of HAGOS. Soccer players constituted 100% of the participants who had filled out Danish



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Table 1 Patient population—age distribution and HAGOS scale scores

HAGOS language version	Danish (N=157)	English (N=146)	Norwegian (N=149)
Age, years, median (IQR)	24 (21–27)	27 (22–30)	22 (19–26)
HAGOS subscale, median (IQR)			
Symptoms	64.3 (53.6–78.6)	62.5 (50.0–71.4)	64.3 (50.0–78.6)
Pain	77.5 (65.0–90.0)	75.0 (62.5–85.0)	72.5 (60.0–85.0)
Activities of daily living	80.0 (65.0–95.0)	80.0 (65.0–90.0)	80.0 (65.0–95.0)
Sport	62.5 (43.8–75.0)	53.1 (37.5–65.6)	50.0 (34.4–65.6)
Physical activities	75.0 (50.0–87.5)	12.5 (0.0–37.5)	25.0 (0.0–50.0)
Quality of life	65.0 (45.0–80.0)	35.0 (25.0–50.0)	35.0 (25.0–50.0)

HAGOS, Hip and Groin Outcome Score.

and Norwegian versions and 29% of the participants who had filled out the English version (table 1). The athletes who had filled out the rest of the English versions were from Rugby (20.5%) and Gaelic football (50.5%).

Hip and Groin Outcome Score

The HAGOS consists of six separately scored subscales: Symptoms (S, Item 1–7); Pain (P, Item 1–10); ADL (A, Item 1–5); Sport/Rec (SP, Item 1–8); PA (PA, Item 1–2) and QOL (Q, Item 1–5). The past week is taken into consideration when answering the items that use Likert scale scoring (0 to 4, 0 indicating no problem). Summative scores are transformed linearly to a 0–100 scale, with zero representing extreme hip and/or groin problems, and 100 representing no hip and/or groin problems. Scores are indirect measurements of underlying latent variables and construct validation testing is a statistical evaluation of the quality of this measurement.

The HAGOS was originally developed in Danish and translated to English and Norwegian and back translated to Danish and checked by the originator of HAGOS (KT) according to the guidelines by Beaton *et al.*¹¹

Modern test theory

We tested validity using the following MTT methods: CFA for ordinal items using weighted least square mean and variance adjusted estimators and an IRT model from the Rasch family, the graphical Rasch model (GRM¹²). We tested measurement invariance in CFA evaluating the fit of CFA models with configural, scalar and partial invariance¹³ and in GRM by testing for DIF.¹⁴ When all items are invariant scores are directly comparable across groups. If they are not, comparison across groups is only possible using conversion tables.

All MTT models rely on the technical requirement of local independence.^{15 16} This requirement can be violated when two items are too similar. If this is the case this can be accounted for by extending the statistical models with correlated error terms (for CFA models) or loglinear interaction parameters (for IRT models). Details about the parametrisation and tests in the GRM are available in online supplemental material.

For the CFA models, fit was evaluated based on the Chi-square test of model fit. This test is the only substantive test of fit available, but it is sensitive to discrepancies from expected values at increasing sample sizes, so for completeness we also evaluated close fit using the root mean square error of approximation (RMSEA), the Comparative Fit Index (CFI) and the Tucker Lewis Index (TLI). For the RMSEA, values below 0.05 and for the CFI and the TLI, values above 0.95 were considered to reflect good fit. Overall GRM fit was evaluated using the Andersen conditional

likelihood ratio test¹⁷ and individual item fit using conditional versions of the INFIT and OUTFIT test statistics.¹⁸ Analyses of item fit produced a p value that tests the null hypothesis of item fit. This means that $p < 0.05$ indicates that the item does not fit. We adjust these p values using the Benjamini-Hochberg¹⁹ procedure to control the false discovery rate at the 5% level. Rasch analysis tested the fit between observed item scores and expected scores. Plots of observed item mean scores against the expected item mean scores across “class intervals” (subsets of respondents with similar subscale scores) were generated. Visual inspection of these provided a test of whether item mean scores increased with domain score. For the CFA models, modification indices were used to add local dependence (correlated error terms), while in the IRT analyses item screening²⁰ was used. Thus, measurement models derived using CFA were confirmed using IRT and vice versa. We added correlation between error terms in a stepwise manner including the ones with the highest improvement in model fit if this improvement was significant (value of modification indices > 3.841) in all language groups.

Initial analyses were conducted for each language version separately and the factor loadings, modification indices, results of the item screening procedure and Rasch item fit statistics were used to choose a model incorporating local dependence. For the model fit indices, factor loadings and item fit statistics were reported. Items where consistent evidence of misfit, and low factor loadings occurred were removed yielding revised HAGOS subscales. Subsequent analyses evaluated measurement invariance across language versions using multiple groups CFA¹³ and confirmed results using GRM’s^{12 21} that incorporate DIF. For the revised HAGOS scales, model fit with configural and scalar invariance were reported, and where scalar invariance was rejected, the fit of models with partial invariance were evaluated. Finally, the implications of reporting revised HAGOS subscale scores were evaluated and conversion tables for scales that were not invariant across language versions discussed.

The CFA models were fitted using the R package lavaan,²² while IRT analyses were performed using the computer program DIGRAM.²³

RESULTS

Psychometric validation of single language versions of HAGOS subscales

Symptoms subscale

For all three language versions, the CFA modification indices pointed to evidence of correlated residuals for the items S6 (‘How severe is your hip and/or groin stiffness after first awakening in the morning?’) and S7 (‘How severe is your hip and/or groin stiffness after sitting, lying or resting later in the day?’). CFA models with correlated error terms showed adequate fit to

Table 2 Fit indices for multiple groups CFA models in original and revised HAGOS scales

HAGOS subscale	Invariance	χ^2	Df	P value	RMSEA (90% CI)	CFI	TLI
S-original	Configural	73.6	36	0.000	0.083 (0.056 to 0.111)	0.986	0.975
	Scalar	138.9	72	0.000	0.079 (0.059 to 0.098)	0.974	0.977
S-revised	Configural	32.9	24	0.107	0.050 (0.000 to 0.088)	0.997	0.994
	Scalar	75.5	52	0.012	0.057 (0.027 to 0.083)	0.990	0.991
	Partial	47.9	34	0.057	0.052 (0.000 to 0.058)	0.997	0.997
P-original	Configural	458.4	105	0.000	0.150 (0.136 to 0.164)	0.940	0.923
	Scalar	385.2	139	0.000	0.109 (0.096 to 0.122)	0.958	0.959
P-revised	Configural	69.9	51	0.041	0.050 (0.011 to 0.077)	0.997	0.995
	Scalar	104.4	81	0.041	0.044 (0.009 to 0.067)	0.996	0.996
	Partial	90.4	73	0.081	0.040 (0.000 to 0.065)	0.997	0.997
A-original	Configural	29.5	15	0.014	0.080 (0.035 to 0.123)	0.995	0.991
	Scalar	53.8	31	0.007	0.070 (0.037 to 0.101)	0.993	0.993
A-revised	Configural	5.2	6	0.524	0.000 (0.000 to 0.098)	1.000	1.000
	Scalar	26.9	20	0.138	0.048 (0.000 to 0.091)	0.998	0.998
SP-original	Configural	308.7	60	0.000	0.166 (0.148 to 0.185)	0.977	0.968
	Scalar	138.9	72	0.000	0.079 (0.059 to 0.098)	0.974	0.977
SP-revised	Configural	25.5	18	0.111	0.053 (0.000 to 0.096)	0.999	0.997
	Scalar	77.5	52	0.012	0.057 (0.027 to 0.083)	0.990	0.991
	Partial	51.6	38	0.069	0.049 (0.000 to 0.080)	0.998	0.998
Q-original	Configural	29.8	15	0.013	0.081 (0.037 to 0.124)	0.995	0.989
	Scalar	76.9	43	0.001	0.073 (0.045 to 0.098)	0.988	0.992
Q-revised	Configural	1.7	3	0.633	0.000 (0.000 to 0.111)	1.000	1.000
	Scalar	33.8	25	0.113	0.048 (0.000 to 0.087)	0.995	0.996

CFA, confirmatory factor analysis; CFI, Comparative Fit Index; HAGOS, Hip and Groin Outcome Score; RMSEA, root mean square error of approximation; TLI, Tucker Lewis Index.

the data for all language versions (online supplemental table 1), but item S2 had a very low factor loading in all samples (online supplemental table 2). Rasch analyses consistently identified local dependence between the items S6 and S7, thus confirming the CFA results. A GRM including this local dependence showed strong evidence of misfit of item S2 ('Do you hear clicking or any other type of noise from your hip and/or groin?') for all language versions (online supplemental table 3). The correlation between S2 and the total scale was lower than expected by the model in all three language versions (online supplemental figure 1). Removing item S2 yielded good model fit for both CFA and IRT models.

Pain subscale

For all three language versions CFA yielded evidence of correlated residuals for the same three item pairs: P3 ('Twisting/pivoting on a weight bearing leg') and P4 ('Walking on an uneven surface'); P6 ('Amount of pain at night while in bed') and P7 ('Amount of pain at night sitting or lying'); P9 ('Walking on a hard surface') and P10 ('Walking on an uneven surface'). This finding was confirmed by the GRM. CFA models that included correlated error terms for these three item pairs showed good fit to the data

for the English and Norwegian versions only, while fit of the GRM was rejected for all versions (online supplemental table 1). The item P2 ('pain in areas other than hip/groin') had low factor loadings (online supplemental table 2) and poor item fit (online supplemental table 3), (online supplemental figure 1). Removing P2 did not greatly improve model fit, but removing both P1 and P2 yielded good CFA and IRT model fit.

ADL subscale

The CFA model had good fit for the Norwegian version, but CFA models in the Danish and English versions did not. Good fit in all CFA analyses was obtained after inclusion of correlated error terms suggested by modification indices (A4 ('Lying in bed (turning over or maintaining the same hip position for a long time)') with A5 ('Heavy domestic duties') for the Danish version and A3 ('Getting in/out of car') with A4 for the English language version), (online supplemental table 1). These models were confirmed by the GRM (online supplemental table 1), but in the Danish and English versions the item A4 had low factor loadings (online supplemental table 1) and some item misfit (online supplemental table 3, online supplemental figure 1). Removing the item A4 yielded models with good fit.

Sports and recreational activities subscale

In both the Danish and English samples, the CFA model fitted very poorly, and modification indices yielded evidence of local dependence for three item pairs. Including local dependence did not yield a model with good fit (online supplemental table 1).

In the Danish sample, findings were not confirmed by the GRM where evidence of local dependence for three other item pairs was disclosed. The overall fit of the GRM was good (online supplemental table 1), but there was evidence of item misfit (online supplemental table 3, online supplemental figure 1).

Table 3 Summary of the five revised HAGOS scales

HAGOS subscale	Removed items	Invariance	DIF items
Symptoms	S2	Partial	S1, S3, S6
Pain	P1, P2	Partial	P3, P5
Activities of daily living	A4	Scalar	None
Sport and recreation	SP1, SP5	Partial	SP2, SP3, SP7, SP8
Quality of life	Q3	Scalar	Nnone

A, activities of daily living; DIF, differential item functioning; HAGOS, Hip and Groin Outcome Score; P, pain; Q, quality of life; S, symptoms; SP, sport and recreation.

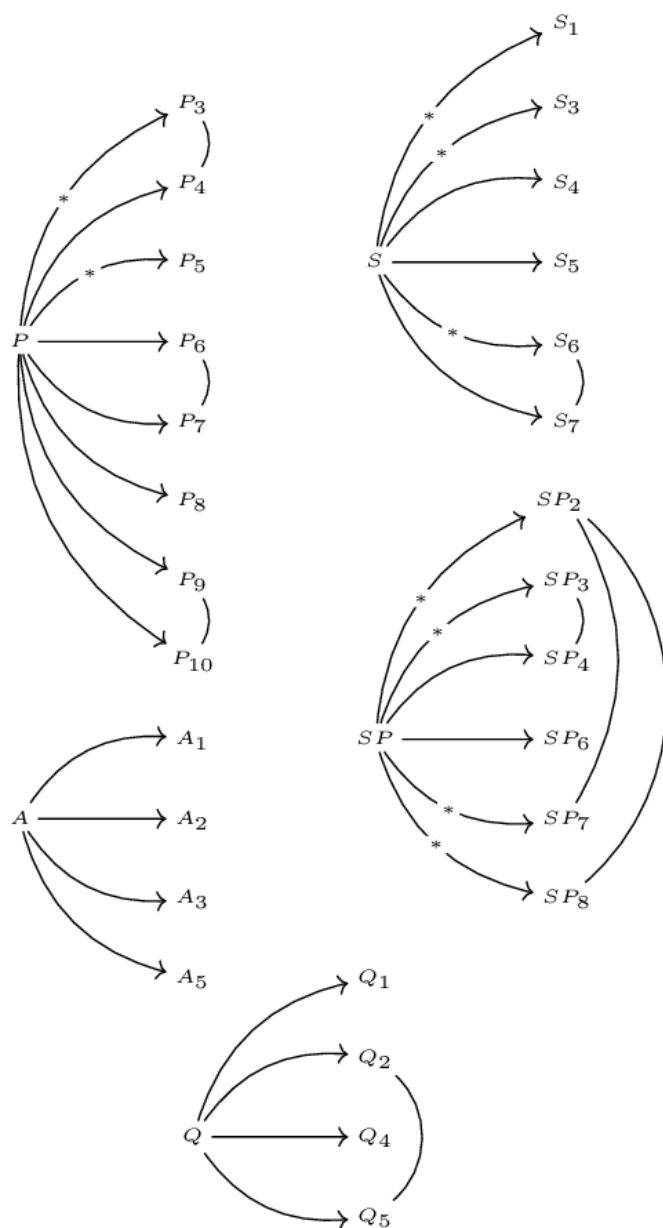


Figure 1 Path diagrams for the five revised HAGOS subscales. *Indicates items that are not invariant across language versions. HAGOS, Hip and Groin Outcome Score.

In the English sample, findings were only partly confirmed by the GRM where significant local dependence occurred for only one of the item pairs. Including local dependence yielded a GRM with acceptable overall (online supplemental table 1), but with evidence of item misfit (online supplemental table 3), (online supplemental figure 1).

In the Norwegian sample, the CFA model fit was not good and modification indices yielded evidence of local dependence for a total of seven-item pairs. Adding all these yielded a CFA model with good fit (online supplemental table 1), but the evidence of local dependence was only partly confirmed by the GRM where only two of the seven-item pairs showed evidence of local dependence and additional evidence of local dependence was disclosed. Including local dependence yielded a GRM with good overall fit (online supplemental table 1), but with evidence of

item misfit (online supplemental table 3), (online supplemental figure 1).

Good model fit was obtained for all three language versions after removal of the misfitting items SP1 and SP5. The graphical evaluation of item fit showed the observed association between the item score and the total score to be lower than expected for these items (online supplemental figure 1), this was most noticeable for SP5. Furthermore, there was evidence of correlated error terms for the same three item pairs SP2 ('Running') and SP7 ('Sudden explosive movements'); SP2 and SP8 ('Situations where the leg is stretched into an outer position'); SP3 ('Twisting/pivoting on a weight bearing leg') and SP4 ('Walking on an uneven surface'). These findings were confirmed by GRMs.

Participation in PA subscale

We did not fit CFA models for single language versions, because the subscale has only two items. The GRM analysis disclosed no evidence of misfit.

QOL subscale

The CFA models showed good fit in Danish and Norwegian version only, while Rasch analysis showed acceptable fit in the English and Norwegian version only. A CFA model with correlated error terms for the items Q2 and Q4 showed good fit for all three versions (online supplemental table 1). In the CFA evidence of local dependence was found for the English version only. There was no significant evidence of item misfit in the Rasch analysis.

Summary of psychometric validation of single language versions

For all HAGOS subscales, except the PA subscale, one or two items were removed in the revised HAGOS. For all revised subscales, CFA analyses identified the same measurement model for all three language versions, and in all cases the GRM confirmed the identified model indicating good fit.

Multiple groups analyses

None of the original HAGOS scales had good fit to models with configural or models with scalar invariance. After removal of the misfitting item S2, all CFA models showed acceptable fit for the revised symptoms subscale. Scalar invariance was confirmed in the GRM ($\chi^2 = 41.0$, $df = 28$, $P = 0.0541$). After removal of the misfitting items P1 and P2, the revised pain subscale fitted all CFA models. Scalar invariance was confirmed in the GRM ($\chi^2 = 77.4$, $df = 58$, $P = 0.0454$). After removal of the misfitting item A4 the revised ADL subscale fitted all CFA models. Scalar invariance was confirmed in the GRM ($\chi^2 = 10.1$, $df = 11$, $P = 0.5191$). After removal of the misfitting items, SP1 and SP5, the revised sports and recreational activities subscale fitted a multiple groups CFA models with configural invariance, but metric and scalar invariance was rejected. A multiple groups CFA model with partial invariance where SP3 and SP8 were not invariant had better fit, and this was confirmed by the Rasch analysis ($\chi^2 = 102.7$, $df = 107$, $P = 0.5984$). Multiple groups CFA models were not fitted for the PA subscale, because the subscale has only two items, but the GRM analysis disclosed no evidence of misfit ($\chi^2 = 9.1$, $df = 7$, $P = 0.2455$). After removal of the misfitting item Q3 the revised QOL subscale showed good fit to multiple groups CFA model with scalar invariance. This scalar invariance was confirmed in the GRM ($\chi^2 = 36.6$, $df = 29$, $P = 0.1554$). Results are summarised in table 2. A summary of the revised HAGOS subscales is reported in table 3 and figure 1.

Table 4 Conversion tables for the revised HAGOS scales

Symptoms			Pain			Sport		
Danish	English	Norwegian	Danish	English	Norwegian	Danish	English	Norwegian
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4.2	2.6	3.2	3.1	3.1	3.1	4.2	4.2	4.2
8.3	5.7	6.2	6.3	6.3	9.3	8.3	8.9	10.3
12.5	8.9	9.2	9.4	6.3	12.2	12.5	13.0	15.1
16.7	12.1	12.5	12.5	8.6	15.2	16.7	17.0	19.4
20.8	15.5	16.3	15.6	11.3	18.2	20.8	21.0	23.4
25.0	19.0	20.7	18.8	14.3	21.2	25.0	25.2	27.4
29.2	22.8	25.3	21.9	17.5	24.2	29.2	29.7	31.2
33.3	26.9	29.9	25.0	20.8	27.3	33.3	34.4	35.0
37.5	31.2	34.6	28.1	24.2	30.4	37.5	39.2	38.9
41.7	35.6	39.2	31.3	27.7	33.4	41.7	44.0	42.8
45.8	40.3	43.7	34.4	31.3	36.4	45.8	48.9	47.0
50.0	45.1	48.2	37.5	34.8	39.4	50.0	53.6	51.3
54.2	50.0	52.6	40.6	38.2	42.4	54.2	58.1	55.6
58.3	55.0	56.9	43.8	41.6	45.3	58.3	62.3	59.9
62.5	59.9	61.1	46.9	44.8	48.3	62.5	66.2	63.8
66.7	64.8	65.1	50.0	48.0	51.3	66.7	69.9	67.6
70.8	69.5	69.0	53.1	51.1	54.3	70.8	73.3	71.1
75.0	74.0	72.9	56.3	54.2	57.3	75.0	76.7	74.5
79.2	78.3	76.7	59.4	57.3	60.4	79.2	80.1	77.9
83.3	82.4	80.5	62.5	60.4	63.5	83.3	83.5	81.5
87.5	86.4	84.5	65.6	63.6	66.6	87.5	87.0	85.4
91.7	90.5	88.9	68.8	66.7	69.8	91.7	90.8	89.9
95.8	94.6	93.7	71.9	70.0	73.0	95.8	95.0	95.2
100.0	100.0	100.0	75.0	73.3	76.3	100.0	100.0	100.0
			78.1	76.6	79.5			
			81.3	80.0	82.8			
			84.4	83.5	86.0			
			87.5	86.9	89.2			
			90.6	90.3	92.1			
			93.8	93.6	94.9			
			96.9	96.8	97.5			
			100.0	100.0	100.0			

HAGOS, Hip and Groin Outcome Score.

For the Symptoms, Pain and Sport subscales the conversion tables reported in [table 4](#) are required.

The English HAGOS questionnaire and the accompanying revised HAGOS excel sheet for scoring each HAGOS subscale (with items not included for subscale scoring marked with yellow) can be found in online supplemental material.

DISCUSSION

The psychometric evaluation of the five HAGOS subscales showed that translated versions were not valid according to MTT, but revised HAGOS subscales with sound psychometric properties can be formed by omitting a single item (for the Symptoms, ADL and QOL subscales) or two items (for the Pain and Sports subscales). Regarding the cross-cultural validity, we found that for three of the five HAGOS subscales values cannot be directly compared across versions. Conversion tables make this comparison possible.

Conversion of HAGOS scores for comparison across different languages and cultures

For HAGOS Symptoms, Pain and Sports and Recreation subscales, a subscale score of for example, 50 points in one

HAGOS language version does not translate to a HAGOS score of 50 points across other language versions, and is illustrated with the following example; This study showed that the Sports and Recreation subscale of the revised HAGOS score, that is, a linear rescaling of the sum of all Sport and Recreation items except SP1 and SP5, is a valid measurement of difficulties with sports and recreation activities. However, for four items there are systematic differences across the three samples. This means that an athlete scoring 50 on Sports and Recreation subscale in the Danish version does not experience the same degree of difficulty as an athlete with the same score on the English version. The graphical presentation in [figure 2](#) shows that typically these differences are not very large, but they can cause bias in multi-country studies or representation between papers published in different populations. The conversion table ([table 4](#)) can be used to translate subscale scores to avoid this. A HAGOS subscale score of 50 on the Danish version would be translated to a score of 51.3 on the Norwegian and 53.6 on the English version.

Methodological considerations and limitations

The results of this study need to be confirmed before the widely used HAGOS PROM is altered. However, this study used three

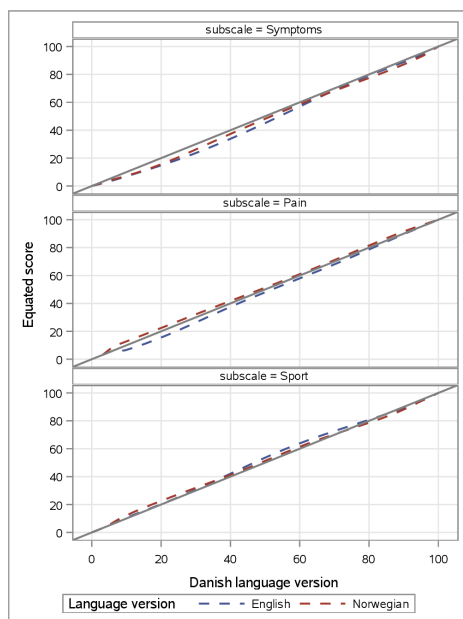


Figure 2 Visualisation of the conversion tables for the three revised HAGOS subscales with differential item functioning using the Danish version as the reference group. HAGOS, Hip and Groin Outcome Score.

separate samples so all disclosed misfits were confirmed in independent data. This suggests that the revised version will also be valid in the other language versions, but validity and invariance for these language versions should of course be confirmed independently. No women were included in the study, so we can not comment on how HAGOS will work in female athletes, and future studies including both men and women, needs to be conducted also in relation to DIF. Furthermore, the use of two different statistical models allowed for cross-validation. It is our suggestion that the items identified as misfitting are retained in the HAGOS questionnaire, but that they are not used for subscale scoring. This serves two purposes: (1) they can be reported as single items, and (2) the original HAGOS subscale scores can be still be computed, for example, for comparison to already published results. The first of these purposes highlights the important point that when an item is misfitting it does not mean that it is inherently bad or irrelevant. It means that the information we get from the item is not adequately reflected by the subscale score. The removal of items SP1 and SP5 from the sport subscale in the revised HAGOS does not mean that ‘squatting’ or ‘running as fast as you can’ are not relevant, but rather that they do not fit well with the other items in the subscale. For this reason, they should be retained as single items. It should be noted that, like the original HAGOS, the revised version has a PA subscale with only two items and hence cannot be validated using CFA. Even though multiple groups CFA and Rasch analysis was feasible and showed excellent fit, the low reliability of a two-item scale remains a limitation of the PA subscale.

PROMS used in sports and orthopedic medicine: implications for the future

An increased awareness of using PROMS with good content validity is seen in orthopaedics and sports medicine, when information on the patient’s and athlete’s perspective is of main priority.^{24 25} For comparison and usage of PROM scores across different languages and cultures, such as in multicentre studies and databases, we recommend that PROMS need to

What are the findings?

- ▶ A revised version of Hip and Groin Outcome Score (HAGOS) and its six subscales using modern test theory provides valid HAGOS subscale measurements in male athletes with groin pain.
- ▶ The revised HAGOS version can be used across the Danish, Norwegian and English language versions of HAGOS.
- ▶ Conversion tables are provided and must be applied to compare HAGOS subscales across Danish, Norwegian and English HAGOS subscale scores. Conversion tables must be used for multicentre randomised controlled trials across different languages.

How might it impact on clinical practice in the future?

- ▶ Valid and cross-culturally adapted Hip and Groin Outcome Score (HAGOS) subscale scores will help clinical practice in relation to male athletes with groin pain to evolve globally.
- ▶ Clinical treatments and their effects in men with groin pain can be better evaluated and compared across different individuals and their language and culture in the future.
- ▶ The revision of HAGOS using modern test theory (MTT) will hopefully inspire and initiate more research in sports and exercise medicine to validate the most commonly used patient reported outcome measures using MTT including: confirmatory factor analysis, item response theory, and Rasch analysis.
- ▶ Clinicians using the HAGOS can continue to use original subscales or use revised subscales along with the omitted item as single items. This is the case when monitoring change in male individuals and when comparing groups. Later studies will have to confirm if the revised version is better at detecting change over time, in women, or between groups. Clinicians who work with Danish and English-speaking teams should ideally use the revised versions and apply conversion tables when comparing scores across language versions.

undergo rigorous psychometric evaluation using MTT,²⁶ such as applied in this study to be of use in this type of research. More research is needed to understand the consequences of the past use of PROMS only developed or validated using CTT, in relation to the implications for previous research findings and the conclusions drawn from these. Since MTT is increasingly used for validation studies the present validation study comes at an appropriate time. To place MTT in context it should be compared with CTT.²⁷

In conclusion, we found that a revised version of HAGOS and its six subscales using MTT provides valid HAGOS subscale measurements in male athletes with groin pain, which means that HAGOS can be used across Danish-speaking, Norwegian-speaking and English-speaking male patients and athletes. In principle, these new subscales have better measurement properties than the original ones, including more precise measurement and better at detecting change.²⁸ Further research in this area is needed, especially in women, to confirm these findings before the instrument is permanently altered. Future research should confirm our findings in independent data and test the validity of the revised HAGOS across patient populations with different age, gender and clinical characteristics.

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